INCH-POUND

MIL-M-14779 (AR)
AMENDMENT 3
23 November 1990
SUPERSEDING
AMENDMENT 2
15 August 1988

MILITARY SPECIFICATION

MAGNESIUM OXIDE, CALCINED
(FOR USE IN AMMUNITION)

This Amendment forms a part of Military Specification MIL-M-14779 (PA) dated 17 May 1968 and is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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1.1 Scope. Delete and substitute as follows:

"This specification covers the requirements for two types of Calcined Magnesium Oxide - one type to be used in Aerial Mines, the other type to be used as an M864 propellant component."

2.1 Add the following under "Specifications":

"MILITARY

MIL-B-117 - Bag, Sleeve and Tubing - Interior Packaging
MIL-B-46506 - Box, Ammunition Packing, Wood, Wirebound
MIL-F-50449 - Filler, Sheet Form (For Use in Ammunition Containers)"

2.2 Other publications. Add the following:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951 - Standard Practice for Commerial Packaging
ASTM-E300-73 - Recommended Practice for Sampling Industrial Chemicals

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)."

Add the following new paragraph:

"CODE OF FEDERAL REGULATIONS

Title 49, Transportation, Parts 100-199"

Add the following new paragraph:

"2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the test of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

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3.1 Requirements. Add the following to the end of the paragraph:

"For use as an XM864 propellant component, the material shall conform to the following requirements:

Property	Min.	Max.	Applicable Paragraph
Magnesium oxide, Calcium basis	88.0	-	4.3.5
Calcium oxide, Calcined basis	-	1.0	4.3.4.3
Weight loss on calcination	-	10.0	4.3.3.
Screen size (USS Sieve) thru No. 325 Sieve	99.9	-	4.3.8
Moisture	-	0.5	4.3.10"

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4.2.2 Sampling. Add the following new paragraph:

* "4.2.2.3 Sampling for M864 propellant component. Approximately 500 grams of the composition shall be selected from each batch to be samples using ASTM Method E300-73 for solids. Samples shall be selected for inspection in accordance with table below. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used.

Number of containers in batch	Number of containers to be sampled
2 - 200	2
Over - 200	5"

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4.3.1.1 Reagents. Delete "anhydrous carbon tetrachloride" and substitute "aliphatic naphtha".

Add the following new paragraph:

"4.3.4.3 Calcium oxide (for M864 propellant component). The calcium oxide shall be determined as follows:

Dissolve 0.630 g of magnesium hydroxide and 0.4 g of magnesium oxide in a mixture of 3 ml of concentrated H_2SO_4 and 22 ml of water. Add 50 ml ethyl alcohol and allow to stand for a minimum of 16 hours. Warm to 50 degrees C and transfer the crystals of calcium sulfate to a Gooch crucible with an asbestos mat that has been washed with a 10% H_2SO_4 solution, water, and ethyl alcohol and has been ignited to constant weight at 800 degrees C. Wash the crystals in the filter with a mixture of 2 volumes of ethyl alcohol and 1 volume of 10% H_2SO_4. Ignite the crucible and contents for 30 min. at 800 degrees C. Cool and weigh.

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Calculations:

$$\text{Wt CaO} = \text{Weigh } \underset{4}{\text{CaSO}} \times 0.412$$

$$\% \text{ Calcium oxide} = \frac{\text{wt CaO determined} \times 100}{\text{wt. sample"}}$$

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Add the following new paragraph:

"4.3.10 Moisture. (For M864 propellant component). Moisture content shall be determined as follows:

Weigh approximately 20 g accurate to 0.1 mg of sample into a wide mouth, glass stoppered, low form drying dish (previously tared). Transfer to an oven at 105 degrees C. Dry for one hour. Remove from oven. Cool in a desiccator and reweigh.

$$\text{Calculation: } \% \text{ moisture} = \frac{(A - B)}{A} \times 100$$

Where:

A = Original sample weight
B = Dried sample"

5.1 Packing. Delete "Packing (see 6.5)" and substitute "Packaging and Packing".

5.1.1 Drums. Delete "Drums" and substitute "Level A". Add the following to the end of the paragraph:

"The drum shall now be placed in a barrier bag constructed from material conforming to Style 1, Type I, Class E, MIL-B-117. The bag shall be heat sealed. The bagged drum shall be placed in a tight fitting, wirebound box conforming to Type I, Class 1, Grade A, MIL-B-46506. The box shall be lined with mandatory fillers conforming to MIL-F-50449. Use additional fillers as required for tight pack."

5.1.2 Level C. Delete in its entirety and substitute the following:

"5.1.2 Level B. Level B packaging shall be the same as Level A, except the barrier bag and mandatory fillers shall be eliminated. However, continue to use fillers as required for tight pack.

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5.1.3 Level C. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with ASTM-D3951".

5.2 Marking. In line 3, delete "Interstate Commerce Commission" and substitute "DOT". In line 4, delete "CFR 71-90" and substitute "CFR 100-199".

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6.1 Intended use. Delete "Mines" and substitute "Mines and M864 propellant".

6.5 Delete paragraph 6.5 in its entirety.

Add the following paragraph:

"6.5 Material sources. Atlantic Solvent No. 49 manufactured by Arco Chemical Co., 260 South Broad St., Philadelphia, PA has been found satisfactory. The aliphatic naphtha (ARCO 49) produced by ARCO has the properties specified in Mellan, I, Handbook of Solvents, Reinhold Publishing Corp., New York."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluated the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 6810-A050)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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